

## **UNIT 1: COMPUTER NETWORK AND COMMUNICATION**

1. Differentiate between guided and unguided media. Provide at least four points of difference and suitable examples.
2. What is network architecture? Explain the main differences between client-server and peer-to-peer network architectures.
3. How does a switch differ from a hub? Explain why modern networks prefer switches over hubs.
4. Compare IPv4 and IPv6 based on address length, representation, and security. Explain why IPv6 was introduced.
5. Explain the differences among the Internet, Intranet, and Extranet in terms of accessibility, purpose, and security, with an example of each.
6. Discuss the advantages and disadvantages of using a ring topology in a computer network.
7. Explain the functions of the following network commands in troubleshooting: ping, ipconfig, tracert, and nslookup.
8. Explain how wireless communication works. Describe the roles and uses of Wi-Fi and Bluetooth technologies.
9. Explain the role of a media converter and an RJ-45 connector in network infrastructure.
10. Explain why broadband is a major improvement over dial-up connections. Describe different types of broadband connections such as DSL, cable, fiber optic, satellite, and wireless.
11. Describe the evolution of communication technologies from 3G to 5G. Compare the generations in terms of speed and data capacity.
12. Differentiate between optical fiber communication and satellite communication in terms of bandwidth and transfer speed.
13. Define PAN, LAN, MAN, and WAN. Describe each type of network with suitable examples.
14. Compare bus and star topologies in terms of cost, performance, and reliability.
15. Compare wired and wireless communication in terms of speed, cost, reliability, and security.
16. Suppose a school wants to set up a network in three separate buildings. Identify a suitable network type and justify how it would support communication, file sharing, and Internet access.
17. Create a network model for a home containing three PCs, one printer, and one mobile device connected to the Internet. Describe the wired/wireless connections and the router, switch, or Wi-Fi technologies used.
18. Design a simple school computer-lab network using at least one switch, ten computers, and Internet access. Explain the design choices and how the devices would communicate.
19. A school network has 200 computers that need unique IP addresses. Explain whether IPv4 or IPv6 would be appropriate and justify your choice with suitable examples.
20. A company wants to share confidential files only with its employees. Identify the appropriate network type and explain why it is suitable.

21. A network suddenly becomes slow. Explain how you would determine whether the problem is related to low bandwidth or low throughput.

## **UNIT 2: DATABASE MANAGEMENT SYSTEM (DBMS)**

1. Define a database. Explain its importance in modern computing systems with suitable examples.
2. Define a Database Management System (DBMS) and explain its major functions.
3. Define primary key and foreign key. Compare them and explain their importance using two linked tables as an example.
4. Explain the structure of a DBMS table by defining rows, columns, fields, and records with a suitable example.
5. Define SQL (Structured Query Language). Explain its importance and role in a DBMS.
6. Explain Data Definition Language (DDL). Describe the syntax and functions of CREATE, ALTER, and DROP with examples.
7. Explain Data Manipulation Language (DML). Describe the syntax and functions of INSERT, SELECT, UPDATE, and DELETE with examples.
8. Explain the main MySQL data-type categories: Numeric, Character/Text, and Date/Time. Give two examples of each.
9. Explain SQL constraints such as NOT NULL, UNIQUE, DEFAULT, and CHECK, and describe how they help maintain data integrity.
10. Explain the three main types of database relationships: One-to-One, One-to-Many, and Many-to-Many, with one real-world example of each.
11. Explain why a DBMS is more advantageous than a simple spreadsheet for managing large amounts of data.
12. Differentiate between a File Processing System and a Database Management System. Give at least two points of difference.
13. Differentiate between a table and a query. Explain the importance of queries in DBMS.
14. What is a report in a database system? Describe its purpose and main features.
15. Explain the relationship among Data, Information, Database, and DBMS with suitable examples.
16. Write SQL queries for a database named CompanyData. Create an Employees table with EmployeeID INT PRIMARY KEY, FirstName VARCHAR(50), LastName VARCHAR(50), and Salary DECIMAL(10,2). Then add Email VARCHAR(100), set Salary default to 25000, rename the table to StaffMembers, insert a sample employee record, retrieve all records, retrieve only FirstName and Salary, increase all salaries by 10%, delete EmployeeID 103, and delete all records while keeping the table structure intact.
17. Suppose a school wants to create a Student table containing StudentID, Name, Class, and Address. Write SQL queries to create the table, insert two records, display students of Class 10, and update the address of a student.

### **UNIT 3: MULTIMEDIA**

- 1.** Differentiate between raster graphics and vector graphics in terms of underlying structure, scalability, and common file formats.
- 2.** Explain how text, images, audio, video, and animation can work together to create an interactive multimedia learning system. Give a practical example.
- 3.** Compare 2D and 3D animation. Explain their dimensional differences and differences in production methods.
- 4.** Explain why data compression is important for multimedia files such as audio, video, and images. Discuss its effect on storage capacity and Internet transmission speed.
- 5.** Explain the significance of frame rate (FPS) and resolution in digital video. Discuss how changing them affects playback quality and file size.
- 6.** Differentiate between MP3, AAC, and WAV audio formats and describe a practical situation for using each.
- 7.** Explain how a video-editing timeline functions. Describe arranging, cutting, trimming, and previewing clips in software such as CapCut.
- 8.** Describe PNG, GIF, JPG, SVG, and PDF file formats and give their typical uses. Explain why PDF is popular in digital communication.
- 9.** Explain the importance of layers in modern graphics-editing software. How do layers help designers organize and safely edit complex composite images?
- 10.** Discuss the role of psychology in animation. Explain how emotions, color theory, and exaggerated visual cues can be used to connect with audiences.
- 11.** What is interactive multimedia? Explain why interaction is important in a multimedia system.
- 12.** Explain the role of multimedia technology in education, business, and entertainment.
- 13.** If you have to create a greeting card, which multimedia components would you use and why?
- 14.** Which type of graphic, raster or vector, would you use for designing a billboard? Justify your answer.
- 15.** Compare the use of multimedia in YouTube and Google Earth. Explain their similarities and differences.
- 16.** Draw a simple multimedia learning-app interface and list the multimedia elements you would include, giving reasons for your choices.
- 17.** Evaluate the impact of video content in comparison with other media forms such as text or audio in digital communication.
- 18.** Describe the main components of multimedia and explain any two uses of multimedia in education.

## **UNIT 4: PROGRAMMING IN PYTHON**

1. Explain the different categories of operators in Python: arithmetic, relational, and logical. Give examples of each.
2. Describe the three main conditional statements in Python: if, if-else, and if-elif-else. Give their syntax and suitable examples.
3. Explain the two types of loops in Python, for and while, with their syntax and a practical example of each.
4. Explain the main Python data types covered in the text: integer, float, string, and boolean. Give an example of each and explain why data types are important.
5. Explain the concept of a user-defined function. Describe parameters, arguments, and return values with suitable code examples.
6. Explain function scope in Python, differentiating between local and global scope, with an example program.
7. Differentiate between void and non-void functions. Explain return values and how a function can return multiple values, with examples.
8. Design a Python program containing at least two user-defined functions: one that takes two numbers and returns their product, and another that takes a list of numbers and prints each number.
9. Explain Python libraries, modules, and packages and their importance in software development. Give suitable examples.
10. Explain the purpose and basic uses of the Math, Random, Pandas, Turtle, and Matplotlib libraries in Python.
11. Describe Turtle graphics in Python and explain its commonly used functions.
12. Discuss syntax errors, runtime errors, and logical errors in Python. Give an example of each and explain their effects.
13. Explain exception handling in Python. Describe the roles of try, except, else, and finally with a suitable example.
14. Explain the different file-opening modes used in Python and give short code examples for read (r), write (w), and append (a) modes.
15. Write a Python script that creates a file named students.txt, writes 'Class 10 Computer Science', then opens the same file in append mode and adds 'Programming in Python'.
16. Write a Python program that reads a file and handles the exception that occurs when the file does not exist.
17. Write the steps and Python code for reading data from a CSV file using Pandas, including importing Pandas and loading the file into a DataFrame.
18. Suppose student\_data.csv contains Name, Age, and Grade. Write a Pandas program to read the file, display the first five rows, calculate the average age, and create a new DataFrame containing only Grade A students.

**19.** Assume marks.csv contains Name, Math, and Computer. Write a Pandas program to read the file and display only students who scored more than 80 in Computer.

**20.** Study the following Pandas program and answer the questions:

```
import pandas as pd
data = {'Name': ['Asha', 'Bikash', 'Chandra'], 'Marks': [45, 72, 60]}
df = pd.DataFrame(data)
print(df[df['Marks'] >= 60])
```

(a) Predict the output. (b) Modify the code to display students who scored less than 60.

**21.** Explain the importance of data visualization in data analysis and business decision-making.

**22.** Explain how to create a basic line chart, bar chart, and pie chart using Matplotlib. Include the import statement, example data, key functions, and display command.

**23.** Differentiate between Matplotlib and Plotly in terms of ease of use, interactivity, customization, and typical visualization features.

**24.** Imagine a Pandas DataFrame containing Month, Category, and Sales for a year. Explain the steps and write Python code using Matplotlib or Plotly to create: (a) a line chart of total monthly sales, (b) a bar graph of annual sales by category, and (c) a pie chart of each category's percentage contribution.

**25.** Write a complete Python program that takes three numbers as input and uses if-elif-else to determine and print the largest number.

**26.** Write a Python program containing calculate\_interest(p, t, r) that takes Principal, Time, and Rate, returns Simple Interest, and prints the result.

**27.** Write a Python program using a while loop to calculate and print the sum of the first 10 even numbers.

**28.** Using Turtle, write a Python program that draws a colored square, using color() to set the pen to blue and fillcolor(), begin\_fill(), and end\_fill() to fill the square with red.

**29.** Write a Python program using the random library as a dice roller that generates a number from 1 to 6 and uses a for loop to roll five times.

**30.** Write a Python program using Matplotlib to draw a bar chart for products = ['A', 'B', 'C'] and sales = [150, 300, 250], including a title and x-axis and y-axis labels.

**31.** Write a user-defined function sum\_of\_digits(num) that uses a while loop to calculate and return the sum of the digits of a multi-digit integer.

**32.** Write a Python program that takes two numbers and prints their sum, difference, product, and quotient.

**33.** Write a Python program that takes a user's name as input and prints a greeting message.

**34.** Write a Python program that takes an integer input and checks whether it is positive, negative, or zero using if-elif-else. If positive, also determine whether it is even or odd using a nested if.

**35.** Write a Python program that accepts two numbers and swaps their values without using a third variable.

**36.** Write a Python program that takes a sentence as input and counts the number of vowels.

**37.** Write a Python program that creates data.txt and writes 'Hello, this is a test file.' into it, then appends 'This is an appended line.'

38. Write a Python program that reads a file and prints its contents to the console.
39. Using the csv module, write a Python program to create students.csv with Name and Marks columns and add three student records. Then write a program to read and display the file contents.
40. Using Pandas, create a DataFrame from a dictionary and save it as a CSV file.
41. Using Pandas and Matplotlib, read data from a CSV file and plot a pie chart for product sales.
42. Write a Python function that takes two numbers and returns their sum.
43. Write a Python function that calculates the area of a circle given the radius as a parameter.
44. Write a Python function that checks whether a number is prime.
45. Write a Python function that takes a list of numbers and returns the maximum number.
46. Write a Python program to calculate the area of a rectangle using a user-defined function.
47. Write a Python program using try-except to handle ValueError when the user enters a non-numeric value.
48. Write a Python program using try-except to handle division by zero and display a custom message such as 'Cannot divide by zero!'.
49. Study the following Turtle program and answer the questions:

```
import turtle
t = turtle.Turtle()
for i in range(4):
    t.forward(100)
    t.right(90)
```

(a) Which shape is drawn? (b) What is the function of right(90)?

50. Study the following program and answer the questions:

```
file = open('test.txt', 'w')
file.write('Hello World')
file.close()
```

(a) In which mode is the file opened? (b) What is the function of write()?

51. Study the following program and answer the questions:

```
import matplotlib.pyplot as plt
months = ['Jan', 'Feb', 'Mar']
sales = [100, 150, 120]
plt.plot(months, sales)
plt.show()
```

(a) Predict the output. (b) Modify it to display a bar graph instead of a line graph.

52. Study a Matplotlib bar-chart program and modify it to display the data as a pie chart instead of a bar chart.
53. Using the Turtle library, write a program with a for loop to draw a hexagon by calculating the correct turn angle dynamically.

## **UNIT 5: ARTIFICIAL INTELLIGENCE (AI) AND CONTEMPORARY TECHNOLOGIES**

1. Explain the concept of Artificial Intelligence (AI) and Machine Learning (ML). Discuss how ML is a subset of AI with suitable examples.
2. Differentiate between supervised and unsupervised learning with suitable practical examples.
3. Describe any four major applications of machine learning in fields such as education, healthcare, or transportation.
4. What are the main types of Artificial Intelligence? Explain them with examples.
5. Explain the five main ways AI helps robots: Perception, Planning, Learning, Communication, and Adaptation, with suitable examples.
6. Explain the role of robotics in modern industries and discuss how robots have transformed healthcare, manufacturing, and education.
7. Describe how AI is used in Microsoft 365 applications such as Word, Excel, and Teams, with examples.
8. Explain the main ethical concerns in AI, focusing on bias, privacy, misinformation, and security. Give examples and ways to reduce or manage each concern.
9. Critically analyze the ethical dilemmas associated with Generative AI tools such as ChatGPT or Midjourney, focusing on copyright infringement, misinformation, and academic dishonesty.
10. Assess the risk of algorithmic bias in AI-driven recruitment tools. Explain how historical hiring data can cause unintended discrimination and propose steps to mitigate it.
11. Define cloud computing and explain any four major services provided by cloud computing platforms.
12. Evaluate the security vulnerabilities of Cloud Computing. Explain why a highly secure national-defense organization might hesitate to use public cloud services and consider a private cloud instead.
13. Discuss how AI, IoT, and Cloud Computing can work together as the foundation of a modern smart city. Explain how the technologies depend on one another.
14. Evaluate the societal impacts of shifting all citizen services to an e-Governance platform. Explain how the digital divide can exclude marginalized communities.
15. Explain how digital governance can make education and healthcare more accessible to citizens, using real-life examples.
16. Evaluate the advantages of e-Education and explain how it has transformed the traditional learning environment for students and teachers.
17. Debate the statement 'e-Education can never fully replace traditional classroom learning' by analyzing psychological, social, and technical limitations of purely online learning for young children.
18. Analyze the advantages and disadvantages of e-Commerce with examples.
19. Formulate a multi-step strategy for a developing country such as Nepal to implement e-Commerce and e-Governance securely while protecting citizens from cyber threats and digital fraud.
20. Explain the concept of the Internet of Things (IoT). Describe two real-world applications of IoT and their impact on society.
21. Discuss the role of IoT in the development of smart cities, with suitable examples.

22. Explain how AI can help in improving cybersecurity.
23. Compare VR, AR, and MR in terms of immersion and interaction with the real world. Give one use case for each and mention one advantage and one limitation of these technologies.
24. Explain Virtual Reality (VR) and Extended Reality (XR). Describe how they can be used in education or entertainment with examples.
25. Analyze how Extended Reality (XR) could transform medicine, including surgical training using immersive VR and diagnostic or surgical assistance using AR overlays.
26. Explain the advantages, disadvantages, and applications of Artificial Intelligence in daily life.
27. A school wants to use AI tools for homework and learning. Prepare responsible-use guidelines covering privacy, honesty, and safe use.
28. Explain the concepts of e-Commerce, e-Governance, and e-Education. Provide one example of each and discuss their importance in modern society.
29. Explain how Machine Learning is used in real-life applications such as spam detection or recommendation systems.
30. Discuss the security and privacy concerns that arise in IoT applications and suggest suitable measures.
31. Explain two differences between traditional/local computing and cloud computing, and discuss advantages and risks of storing data in cloud servers instead of local computers.
32. Explain how cloud computing can support online learning and remote work or collaboration among team members.
33. Discuss the potential economic impacts of advanced robotics and AI in manufacturing, including both job creation and job displacement.
34. Analyze the ethical and legal complexities of responsibility when an autonomous self-driving car causes a fatal accident, considering the driver, programmer, and manufacturer.